

# Surya Bansal

✉2647surya@gmail.com | 📞+91-8826123035 | 🔗Linkedin | 🐙Github

## EXPERIENCE

### JPMORGAN CHASE & CO. 📄 | Bangalore, IN

#### SOFTWARE ENGINEER II

June 24 – Present

- Architected and built **low-latency risk publishing systems** for commodities trading, processing **\$5B+ in positions** and scaling across **multiple front-office trading desks**.
- Led the modernization of a **petabyte-scale** end-of-day publication platform, delivering **1+ PB** of data to downstream systems with **99.9%+ SLA compliance**; introduced **distributed delivery, caching, and horizontal scaling** to cut publication time by **50%** while maintaining cost efficiency at scale.
- Extended the risk publishing platform to support **new systematic trading workflows and downstream feeds**, integrating **GPU-based risk models** to enable expanded trading coverage.
- Owned the **end-to-end architecture and development** of a fine-grained, tenant-based, tiered **reconciliation platform supporting multiple JPMorgan businesses**, reconciling **petabyte-scale datasets** in under 1 hour and reducing **manual production verification by 80%**.
- Built **LLM-powered incident-resolution workflows** combining retrieval and tool invocation for automated root-cause analysis and production support, reducing **issue-resolution time by 70%**.
- Developed **MCP servers and Jira/Confluence workflow automations** to integrate engineering utilities with AI agents, streamlining **incident documentation, problem tracking, and developer workflows**.

### WALMART 📄 | Bangalore, IN

#### SOFTWARE ENGINEER

July 22 – June 24

- Developed a **PII Detection Service** from scratch using the Microsoft Presidio framework, enabling domain-based routing and delivering highly accurate, customized redaction results, saving the firm **\$5M annually**.
- Onboarded multiple tenants with Nginx reverse proxy servers and scaled the service through data preprocessing, achieving a **90% reduction in downtime** in the PII service.
- Developed a metrics dashboard using **Kafka and GCP BigQuery** to monitor Micro Frontend services, significantly enhancing real-time data analysis and improving operational efficiency.
- Engineered a redaction system automating legal document processing, delivering **\$2M in annual savings** with an **80% reduction in manual redaction** time.
- Owned the end-to-end development**, testing, deployment, and maintenance of the Soteria Redaction Platform. Evaluated cost-effective enterprise licenses and partnered with cross-functional teams to drive platform adoption across the organization.
- Built a high-throughput notification microservice for One Walmart serving over **100K daily impressions**, implementing caching, asynchronous processing and performance tuning for low-latency delivery..
- Built **GenAI prototypes** using prompt engineering, LLM APIs, and enterprise workflows for document summarization and intelligent automation, expanding AI adoption across the team.

### FAMPAY (YC S19) 📄 | Bangalore, IN

#### SOFTWARE ENGINEERING INTERN

Dec 21 – June 22

- Contributed to scaling FamPay's (YC S19, Sequoia-backed) digital payments infrastructure.
- Developed an automated reward distribution system for **5M+ users** that increased efficiency in customer engagement.
- Enhanced the overall test suite and built restful APIs using Redis cache, Celery and Django REST.

### RACCHABANDA 📄 | San Diego(Remote), CA

#### SOFTWARE ENGINEERING INTERN

Aug 21 - Dec 21

- Developed a web conferencing app using React, Redux, Node.js and optimised video latency.
- Implemented **virtual and blur backgrounds** with Tensorflow.js model(bodypix) using Web Workers.
- Added support for type ORM to SQLite APIs by migrating to Prisma.

## PROJECTS

### CROP RECOMMENDATION APP 📄

Nov 24 | Grace Hopper Sustainability AI Hackathon (Winner)

- Built an AI-driven mobile app that combined weather, soil, and market data to help farmers make informed crop choices and improve yield outcomes.
- Stack: React Native, Generative AI models using Llama, Mistral AI 7B Instruct model, OpenAI, FastAPI, Data Sources like Weather API, Soil API, Yield API for real-time data

### SEARCH AI TOOL

April 24 | Chalaang 2.0 Hackathon (Top 5)

- Engineered a context-aware search solution powered by RAG, LLMs, and vector stores, improving the accuracy of answers retrieved from the web.

### PR REVIEW TOOL

Oct 23 | Walmart X Google GenAI Hackathon (Winner)

- Developed a PR review bot to automate the code review process by analyzing pull requests. Used technologies like Prompt Engineering, Google Vertex AI, and GenAI models: textbison, codebison.

### SMART CITY MOBILE APPLICATION 📄

Sept 20 | The Global Hack (Winner)

- Developed a React Native mobile application, enabling users to give direct feedback to the local government about their health, sanitation, and social distancing in the area during COVID-19.
- Awarded a \$300 MIT Solve micro-grant (2021) and €5,000 under hackathon's E-Governance track (2020).
- Represented India and built a smart city MVP which was mentioned by the Ministry of Electronics and Information Technology and The Economic Times.

### REACT WHITEBOARD APPLICATION 📄

April 21 | Medium Publication

- Built a mobile-first responsive whiteboard using React Hooks, Redux, and Canvas API, and authored a detailed Medium tutorial with 40k+ interactions.

## TECHNICAL SKILLS

### LANGUAGES & FRAMEWORKS

• Python • C++ • TypeScript • JavaScript  
• Lua • Node.js • Next.js

### DISTRIBUTED SYSTEMS & CLOUD

• Kafka • GraphQL • Microservices  
• Distributed Systems • Redis • PostgreSQL  
• AWS • GCP • Azure • Kubernetes • Docker  
• Linux • Grafana • Prometheus

### DATA, AI & TOOLS

• BigQuery • NumPy • LangChain  
• RAG • Vector Databases • Git • Postman

## EDUCATION

### GURU GOBIND SINGH

### INDRAPRASTHA UNIVERSITY, DELHI

B.Tech in Information Technology

2022 | CGPA: 9.08